Dr. THU ZAR NWE
Postdoctoral researcher, Plant Ecology
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ResearchGate, Academic website
Nationality: Myanmar

PROFESSIONAL PROFILE

Ecologist with a PhD in Ecology and Evolution and experience in natural and experimental grassland ecosystems, plant functional traits, soil biodiversity and plant-soil interactions. Experienced in ecological field and laboratory research, including independently designing a trait-based litter-bag experiment across plant functional groups. Skilled in ecological data using R, including mixed-effects models and structural equation modelling.

RESEARCH INTERESTS

Plant-soil interactions | Soil biodiversity and ecosystem functioning | Grassland ecology | Functional traits of soil fauna | Collembola | mites | Collembola mouth-part traits and feeding ecology | Trophic relationships | Soil decomposition processes | Biodiversity-multifunctionality relationships

EDUCATION

Doctor of Philosophy (Ph.D.), Ecology and Evolution
Institute of Plant Sciences (IPS), University of Bern, Switzerland, Sep 2017 - Aug 2023
PhD research: Conducted within the large-scale PaNDiv grassland experiment, combining field and laboratory approaches to investigate direct and indirect effects of nitrogen enrichment, plant diversity and fungal pathogens on plant-soil interactions, soil organisms and ecosystem processes.

Master of Research (M.Res.), Botany
Department of Botany, Yadanabon University, Myanmar, Jun 2010 - Mar 2011
M.Res. thesis: Conducted a field-based survey of algal diversity and water quality in the Sedaw Lay Diversion Dam, Myanmar. Collected and identified 78 algal species across major taxonomic groups, documented specimens photographically, and assessed physicochemical water properties including dissolved oxygen, BOD, pH, hardness, alkalinity and major ions.

Master of Science (M.Sc.), Botany
Department of Botany, Yadanabon University, Myanmar, Jun 2008 - Mar 2010

Bachelor of Science (B.Sc., Honours), Botany
Department of Botany, Yadanabon University, Myanmar, Dec 2003 - Sep 2007

TRAINING

- Soil Collembola identification and trait measurement internship - Soil Functional Ecology Lab, Université du Québec à Montréal (UQAM), Canada (Mar-May 2019)
- Soil nematode identification internship - Department of Ecology, Swedish University of Agricultural Sciences (SLU), Uppsala, Sweden (Oct 2018)

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher, Plant Ecology
Institute of Plant Sciences (IPS), University of Bern, Switzerland, Jun 2024 - Present
- Effects of fertilization on Collembola (Springtail) abundance in experimental and natural grasslands (PaNDiv and Biodiversity Exploratories).
- Impact of land-use intensification on the composition and diversity of plants, Collembola, and fungi and their interactions: a time-series study (Biodiversity Exploratories).
- Interactions of collembolan and plant communities: comparing experimental grasslands and real-world grasslands (PaNDiv and Biodiversity Exploratories).

- Independent project: Leaf Calcium Variation Shapes Collembola Community and Functional Traits in Grassland Ecosystem, designed a trait-based litter-bag experiment across plant functional groups, conducted microarthropod extraction, Collembola identification and mouthpart trait measurement, and trophic groups of mites.

Research Assistant, Community Ecology
Institute of Plant Sciences (IPS), University of Bern, Switzerland, Sep 2023 - Mar 2024

- Investigated decomposition rates using bait-lamina strips with home and away litters, conducted microarthropod extraction, trait measurement, and feeding-activity assessment across litter resources.

Research Assistant, Terrestrial Ecology
Institute of Ecology and Evolution (IEE), University of Bern, Switzerland, Sep 2023 - Feb 2024

- Identified Collembola species from the HOME (Hasli Outdoor Mesocosm Experiment) grassland experiment and measured body size as a functional trait and contributed data for research on soil fauna responses to heatwave and drought treatments.

Teaching Experience, Department of Botany

- Lecturer, Taunggyi University, Myanmar (on leave for study) - Dec 2019-May 2021
- Assistant Lecturer, Mandalay University, Myanmar (on leave for study) - Sep 2017-Dec 2019
- Assistant Lecturer (Full-time), Mandalay University, Myanmar - Aug 2016-Aug 2017
- Demonstrator (Full-time), Kyaukse University, Myanmar - Sep 2012-Aug 2016

GRANTS & SCHOLARSHIPS

- Scholar At Risk Grant, postdoctoral researcher program, University of Bern (Jan 2025 - Dec 2026)
- Scholarship from Scholar At Risk for PhD program, University of Bern (Jan 2022 - Dec 2022)
- Swiss Government Excellence Scholarship for PhD program, University of Bern (Sep 2017 - Dec 2021)
- TCS Colombo Plan Scholarship, NITTTR, Chennai, India (Dec 2014 - Jan 2015)

PUBLICATIONS

Nwe, T.Z., Maaroufi, N., Allan, E., Soliveres, S., Kempel, A. (2023). Plant attributes interact with fungal pathogens and nitrogen addition to drive soil enzymatic activities and their temporal variation. Functional Ecology, 37(3), 564-575.

Pichon, N.A., Cappelli, S.L., Soliveres, S., Mannall, T., **Nwe, T.Z.**, Hölzel, N., Klaus, V.H., Kleinebecker, T., Vincent, H., Allan, E., (2024). Nitrogen availability and plant functional composition modify biodiversity-multifunctionality relationships. Ecol. Lett. ele.14361.

Manuscripts in submission/preparation:

- **Nwe, T.Z.**, Handa, T., Kempel, A., Soliveres, S., Maaroufi, N., Raymond-Léonard, L.J., Pichon, N.A., Cappelli, S.L., Allan, E. Collembolan functional composition is driven by plant attributes rather than nitrogen enrichment and fungal pathogens. (in submission)
- **Nwe, T.Z.**, Pichon, N.A., Cappelli, S.L., Kempel, A., Soliveres, S., Allan, E., Maaroufi, N. Leaf traits and plant cover mediate the impacts of nitrogen addition and plant diversity on soil mites. (in prep)
- **Nwe, T.Z.**, Allan, E., Fischer, M. Direct and indirect effects of fertilization on Collembola (Springtail) abundance differ in natural and experimental grasslands. (in prep)
- Liu, X., Godoy, O., Daniel, C., Pichon, N.A., Cappelli, S.L., Mannall, T., **Nwe, T.Z.**, Vincent, H., Neubauer, L.C., Maaroufi, N.I., Allan, E. Global change alters how niche differences translate into multiple ecosystem functions. (in prep)

Additional early-career publications in algal taxonomy and microbiology (2014-2017) available on request.

ACADEMIC SUPERVISION

- Main supervisor, 3 Bachelor research practicals, IPS, University of Bern - pesticide effects on Collembola/mites, soil nematode responses to pathogen removal, soil mite functional groups vs. plant diversity (2023-2024)

- Main supervisor, 2 Bachelor group projects on nitrogen/fertilization effects on decomposition, Advanced Community Ecology course, University of Bern (2022-2023)
- Co-supervisor, 1 Bachelor theses, IPS, University of Bern - home-field advantage in decomposition, plant-soil feedbacks of temperate tree species (2024-2025)

OTHER PROFESSIONAL ACTIVITY AND COLLABORATION

Peer reviewer: Functional Ecology (2023).

Field collaborator: Research for Development (r4d) project on participatory land-use modelling, Centre for Development and Environment, Myanmar (2016).

SELECTED PRESENTATIONS & POSTERS

- Direct and indirect effects of grassland fertilization on Collembola communities, Biology26, Neuchâtel, Switzerland (Feb 2026)
- How do changes in agricultural treatments impact biodiversity? (Poster), Swiss Global Change Day, Bern (Apr 2023)
- Impacts of nitrogen addition and plant diversity on soil fauna in grassland ecosystems, Biology23, Geneva (Feb 2023)
- Drivers of Collembola community and their functioning, INTECOL, Geneva (Aug 2022)
- Effect of nitrogen enrichment, plant richness and fungal pathogens on soil enzyme activities, British Ecological Society, Online (Dec 2020)
- Direct and indirect effects of nitrogen on functional traits of soil fauna (Poster), Biology20, Fribourg, Switzerland (Feb 2020)
- Direct and indirect effects of nitrogen on functional traits of Collembola (Poster), British Ecological Society, Belfast, Northern Ireland (Dec 2019)

TECHNICAL SKILLS

Data & Analysis: R programming, linear mixed-effects models, structural equation modelling, statistical data analysis.

Laboratory & Field: Soil enzymatic activity measurement (microplate assays with spectrophotometer), microbiological culturing and assessment of microbial activities, Collembola and algae identification, trophic group assessment of mites/nematodes, soil fauna trait measurement, grassland field sampling and biodiversity assessment.

OUTREACH & VOLUNTEERING

- Co-organizer, Plant Ecology lab retreats, University of Bern (2025-2026)
- Science outreach participation, Ancienne Gare, Fribourg, Switzerland (2020)
- Online volunteer teaching for Myanmar students (May 2026)